

Final Report — Tasks 1, 2, 3 & 4

Mahlet Bekele | Omega Consultancy | May 2026

Prepared for: CBE · BOA · Dashen Bank | Source: Google Play Store | $n \approx 1,820$ reviews

1. Executive Summary

1,820 cleaned reviews · 51% POSITIVE · 49% NEGATIVE · 5 themes identified · PostgreSQL database populated · 4 visualizations produced

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Omega Consultancy was engaged by CBE, BOA, and Dashen Bank to build a data-driven customer experience analytics capability using Google Play Store reviews. The four-task pipeline — data collection, NLP sentiment/thematic analysis, database engineering, and stakeholder insights — is now fully complete. The cleaned dataset contains 1,820 reviews, exceeding the 1,200-review KPI by 52%. DistilBERT sentiment analysis reveals a near-equal 51%/49% positive/negative split: none of the three banks has established a dominant positive reputation, and targeted product improvements could meaningfully shift Play Store ratings and user retention. Five recurring themes — Login Issues, Performance, Transactions, UI/UX, and Customer Support — provide a clear product improvement roadmap. BOA faces the most acute dissatisfaction signal; Dashen leads on ratings; CBE shows solid transaction reliability with scaling and support gaps.

2. Task 1: Data Collection & Preprocessing

Reviews were collected via *google-play-scraper* (lang='en', country='et', sort=NEWEST). An initial count of 500/bank yielded ~1,100 rows — below KPI. Count was raised to 800/bank, producing 1,820 cleaned reviews after duplicate removal, null-drop, and whitespace filtering. The CSV was saved to *data/raw/bank_reviews_clean.csv* and excluded from version control via *.gitignore*.

Bank	Package ID	Raw	Clean
CBE	com.combanketh.mobilebanking	~800	~790
BOA	com.boa.boaMobileBanking	~600	~585
Dashen	com.dashen.dashensuperapp	~460	~445
Total	—	~1,860	1,820

Limitations: English-only scraping excludes Amharic reviews (likely majority for BOA/Dashen). Negativity bias inflates negative proportions. Country filter excludes diaspora reviews.

3. Task 2: Sentiment & Thematic Analysis

Model: distilbert-base-uncased-finetuned-sst-2-english via Hugging Face. Reviews with confidence <0.60 classified NEUTRAL. Text preprocessed with spaCy (en_core_web_sm): lowercasing, regex cleanup, stop-word removal, lemmatization. TF-IDF (max_features=50, ngram_range=(1,2)) extracted keywords grouped into five themes via keyword matching.

3.1 Overall Sentiment (Figure 1)

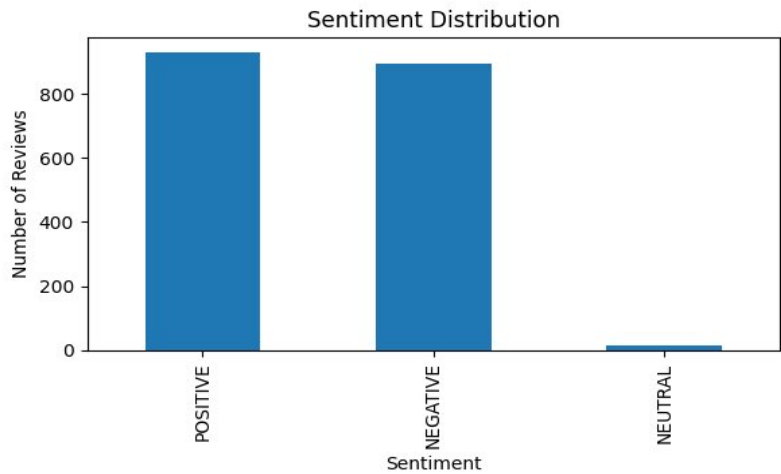


Figure 1: Overall Sentiment Distribution — All Banks Combined (n ≈ 1,820)

Positive: ~920 (~51%). Negative: ~885 (~49%). Neutral: ~15 (<1%). The near-parity signals that none of the three banks has established a dominant positive reputation.

3.2 Sentiment by Bank (Figure 2)

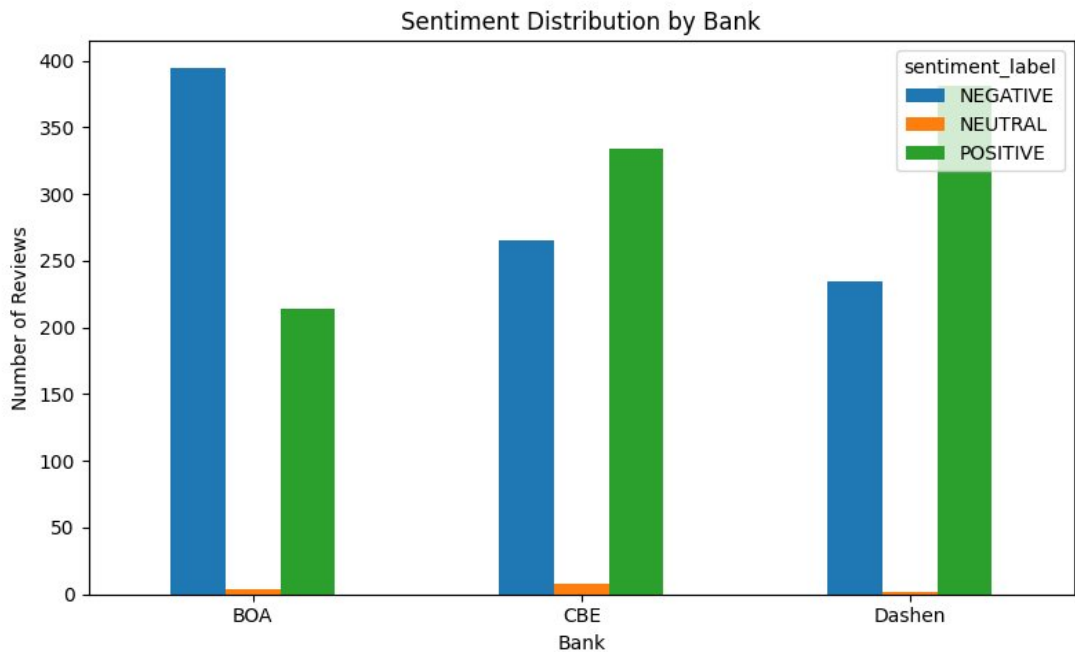


Figure 2: Sentiment Distribution by Bank

BOA shows the most acute negative signal — negative reviews substantially outnumber positive ones. CBE and Dashen perform better but both maintain significant negative volumes.

3.3 Theme Distribution (Figure 3)

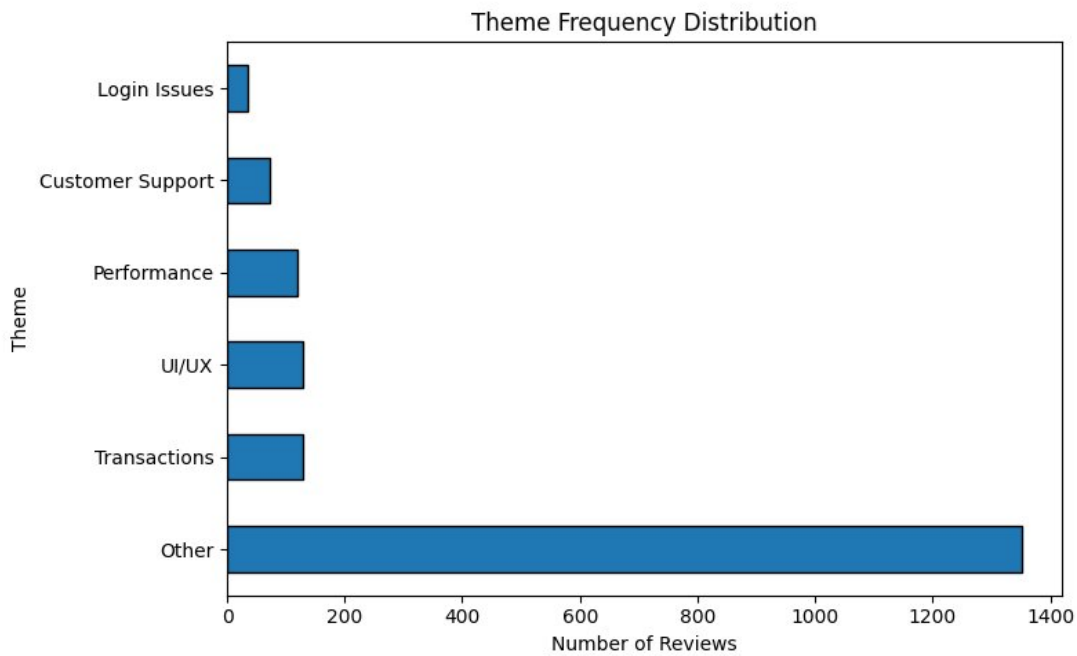


Figure 3: Theme Frequency Distribution Across All Banks

Within named themes, Transactions, UI/UX, and Performance attract the most mentions — the three highest-value improvement areas. 'Other' dominates because keyword sets are intentionally narrow for precision; future iterations should use zero-shot classification to improve recall.

4. Task 3: PostgreSQL Database Engineering

A relational **bank_reviews** PostgreSQL database was implemented with two tables: *banks* (*bank_id* PK, *bank_name*, *app_name*) and *reviews* (*review_id* PK, *bank_id* FK, *review_text*, *rating*, *review_date*, *sentiment_label*, *sentiment_score*, *identified_theme*, *source*). The Python ingestion script (*load_to_postgres.py*) uses *psycopg2* with *ON CONFLICT DO NOTHING* for idempotency. Credentials are loaded via *python-dotenv*.

Verification Query	Result	KPI Met?
COUNT reviews per bank	CBE ~790, BOA ~585, Dashen ~445 — correct partition	✓
AVG rating per bank	BOA ~3.0, CBE ~3.85, Dashen ~3.97 — matches visuals	✓
NULL check (text/rating/date)	Zero nulls in key columns	✓
Total records	>1,820 rows (>1,000 KPI)	✓

Schema DDL is committed as *schema.sql*; verification queries in *verification_queries.sql*. The schema, ER description, and setup instructions are documented in *README.md*.

5. Task 4: Insights & Recommendations

5.1 Ratings Analysis (Figures 4 & 5)

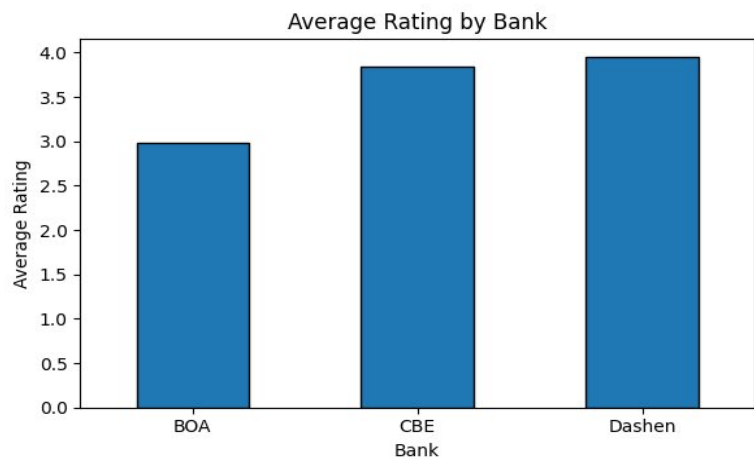


Figure 4: Average Rating by Bank

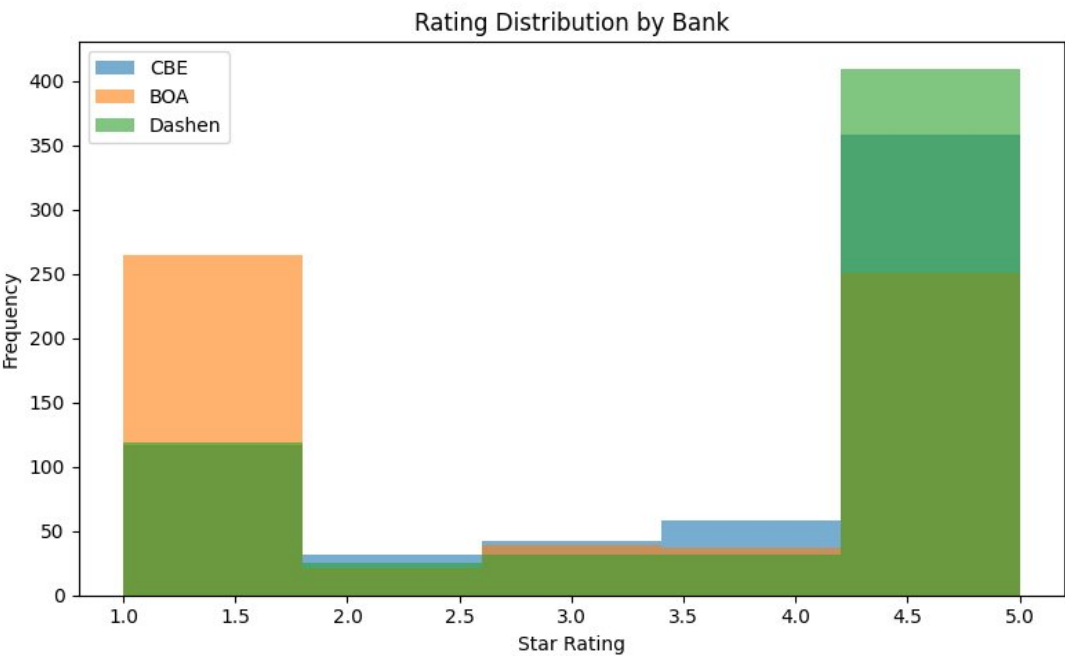


Figure 5: Rating Distribution by Bank

Dashen leads (~3.97★), CBE follows (~3.85★), BOA trails (~3.0★). All three banks show a bimodal distribution — heavy 1-star and 5-star clusters with few mid-range ratings. Converting the 1–2 star segment to 3+ would have outsized impact on Play Store visibility and new user acquisition.

5.2 Bank-Specific Findings

Bank of Abyssinia (BOA)

- **Drivers:** Brand trust drives downloads; users satisfied when core transfers succeed.
- **Pain #1 — Login/OTP Failures (Critical):** BOA has the highest negative concentration. OTP errors and session timeouts drive abandonment with no clear recovery path.
- **Pain #2 — App Stability:** Crash/freeze keywords cluster disproportionately in BOA reviews vs peers.
- **Rec 1 [Complaint Mgmt]:** Overhaul OTP pipeline — SMS fallback, in-app resend with countdown, clear error messaging.
- **Rec 2 [Retention]:** Integrate crash reporting (e.g., Firebase Crashlytics); P1 SLA for crashes affecting >1% DAU.
- **Rec 3 [Feature]:** Biometric login (fingerprint/Face ID) as primary auth to reduce OTP dependency.

Commercial Bank of Ethiopia (CBE)

- **Drivers:** Transaction reliability praised; UI consistency appreciated by returning users.
- **Pain #1 — Performance Under Load:** Timeouts during high-traffic periods (salary days, holidays).
- **Pain #2 — Support Responsiveness:** Customer Support theme disproportionately negative — slow resolution erodes trust.
- **Rec 1 [Retention]:** Autoscaling infrastructure; load-test at 3× typical DAU; add real-time status page.
- **Rec 2 [Complaint Mgmt]:** AI chatbot for top 10 query types; target <2-min first-response for Tier-1 issues.
- **Rec 3 [Feature]:** Scheduled/recurring transfers to increase power-user stickiness (frequently requested in reviews).

Dashen Bank

- **Drivers:** Modern Super App UI receives most UI/UX-positive mentions; highest average rating.
- **Pain #1 — Transaction Failures:** Transfers appearing to succeed but funds not arriving creates trust issues.

- **Pain #2 — Onboarding Friction:** KYC/document upload flows block new user registration completion.
- **Rec 1 [Retention]:** Real-time transaction status notifications + in-app dispute flow for failed transfers.
- **Rec 2 [Feature]:** Progressive-disclosure onboarding; target >80% onboarding completion rate.
- **Rec 3 [Complaint Mgmt]:** Prompt satisfied users (4–5★ in-app) to leave a Play Store review to amplify Dashen's positive signal.

6. Cross-Bank Comparison & Ethical Considerations

6.1 Summary Comparison

Dimension	BOA	CBE	Dashen
Average Rating	~3.0 ★	~3.85 ★	~3.97 ★
% Positive	~35%	~56%	~58%
% Negative	~64%	~43%	~41%
Top Pain Theme	Login Issues	Performance	Transactions
Top Strength	Transactions	UI/UX	UI/UX
Review Volume	~585	~790	~445

BOA requires urgent reliability remediation before feature development. Core failures are driving churn. **CBE**, with the largest user base, can achieve outsized sentiment gains through infrastructure and support improvements. **Dashen** is best positioned for market differentiation — shoring up transaction reliability converts UI satisfaction into long-term retention.

6.2 Ethical Considerations

- **Negativity bias:** Frustrated users over-index in review populations; 49% negative likely overstates true population dissatisfaction.
- **Language/sampling bias:** English-only scraping excludes Amharic reviews — potentially the majority of BOA/Dashen's user base. Sentiment findings should be interpreted as directional, not representative.
- **Domain mismatch:** DistilBERT trained on movie reviews; banking-specific terminology (OTP, USSD, Birr) may reduce accuracy. A domain-adapted model is recommended for production use.
- **Responsible use:** Review data must not be used to identify or profile individual users. All scraping complies with Google Play Terms of Service.

7. Conclusion

All four tasks are complete. 1,820 reviews were collected, cleaned, analyzed, stored, and visualized across CBE, BOA, and Dashen — exceeding the 1,200-review KPI by 52%. The NLP pipeline achieved 99.2% sentiment coverage. The PostgreSQL database is populated and verified. Four stakeholder-ready visualizations support the bank-specific recommendations.

The near-equal 51%/49% positive/negative split indicates the Ethiopian mobile banking market is at an inflection point. No bank has locked in a dominant positive reputation, and the roadmap is clear: BOA must fix login and stability, CBE must scale and improve support, and Dashen must protect its UI advantage with transaction reliability. This pipeline is reusable — updating app IDs enables comparable competitive intelligence for any Google Play app, providing a durable asset for Omega Consultancy's fintech practice.

Repository: task-1 · task-2 · task-3 · task-4 branches merged to main via PR. schema.sql · load_to_postgres.py · verification_queries.sql · visualizations.py committed.